VisionMate

Literature Review

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Background and Motivation

Evolution of Assistive Technologies

Response Latency

High response latency remains a critical issue in assistive technologies, often hindering real-time user interactions and frustrating the visually impaired who rely on immediate feedback.

Internet Dependency

Many current applications heavily depend on internet connectivity, limiting their usability in areas with poor reception, which poses significant challenges for users in remote locations.

Offline Use

Limited offline capabilities restrict the functionality of assistive applications, making them unreliable in situations where internet access is unavailable, thus undermining their effectiveness in daily use.

Object Detection on Mobile

The effectiveness of object detection on mobile devices hinges on several key factors:

- Balancing accuracy and latency
- Energy efficiency considerations
- Utilizing frameworks like TensorFlow Lite and MobileNet-SSD
- Vision Mate aims for ≥85% accuracy with a response time of ≤2 seconds.

OCR Technologies

OCR advancements enhance accessibility for visually impaired users. Key aspects include:

- Overview of Tesseract OCR
- Advantages of Google ML Kit
- Support for multiple languages
- Integration of OCR with real-time TTS to improve user experience in Vision Mate.

Research Gaps in Assistive Technologies

System Limitations

Current systems face heavy cloud reliance issues.

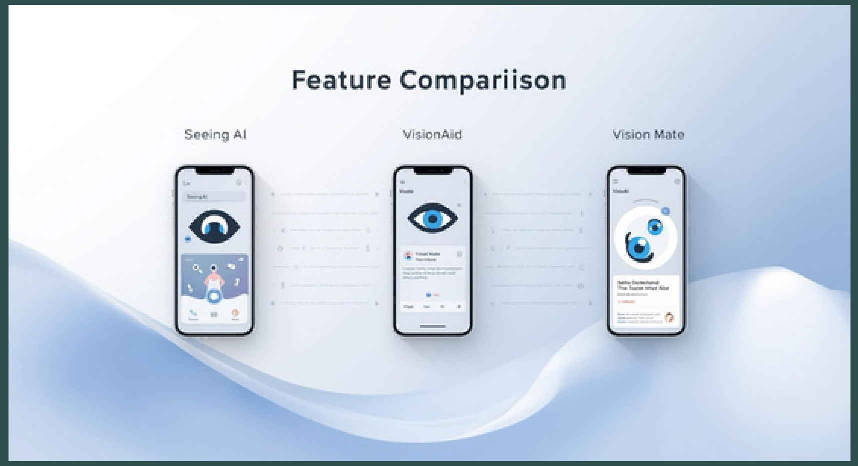
Real-Time Constraints

Many systems lack real-time processing capabilities.

Accessibility Focus

Insufficient emphasis on user-friendly interfaces exists.

Comparative Feature Analysis of Assistive Applications



Object Detection

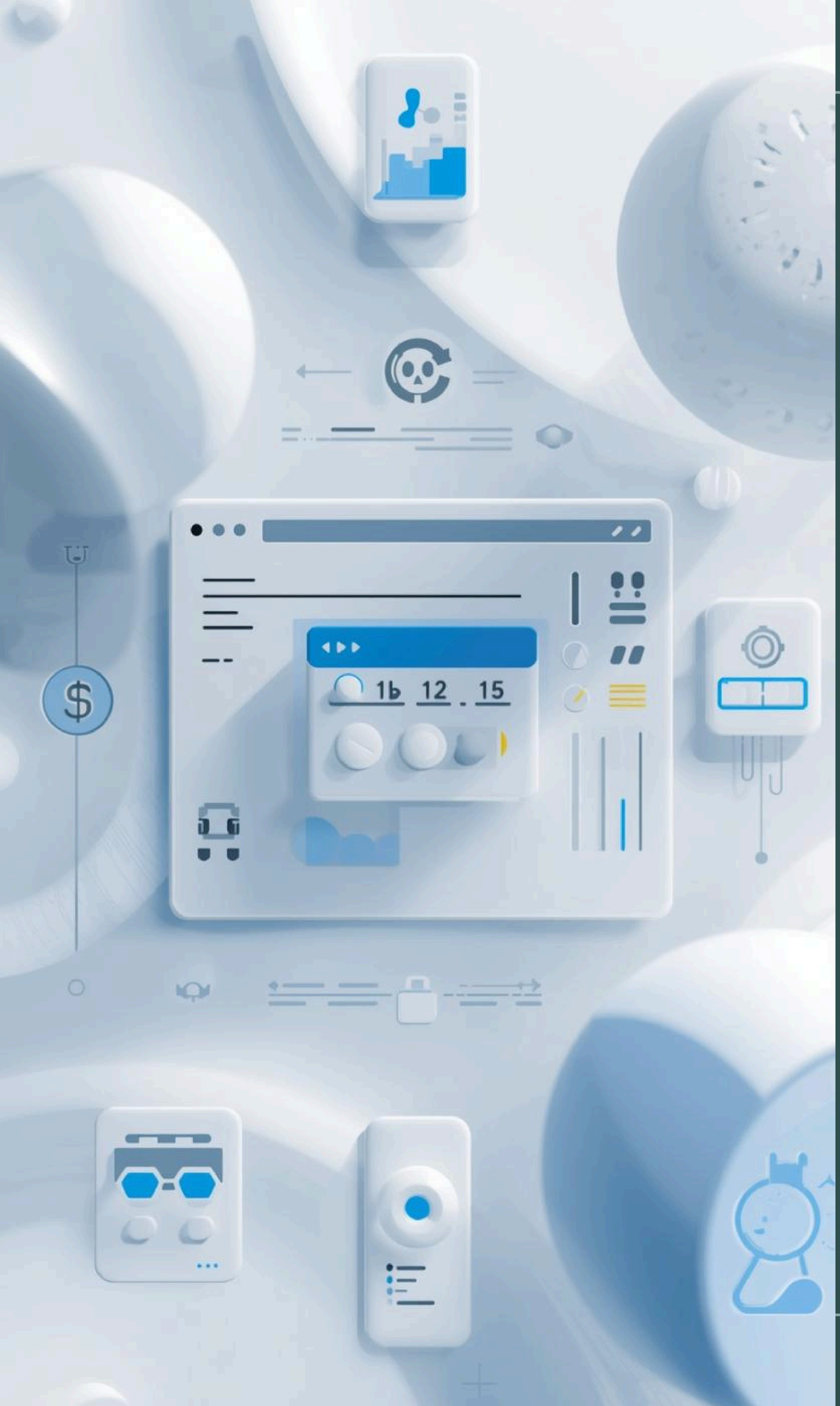
Vision Mate achieves superior accuracy in detection tasks.

OCR Technology

Robust OCR integration enhances accessibility for users.

User Accessibility

Vision Mate prioritizes an intuitive and accessible interface.



Vision Mate KPIs

01 Focus on offline on-device processing for reliability.

02 Response time target is set to ≤ 2 seconds.

03 Accuracy KPIs aim for $\geq 85\%$ detection and $\geq 90\%$ OCR.